Grade 10 Physical Science Memo
Paper 1 November 2025

Question 1 – Multiple Choice

1.1 A
1.2 C
1.3 C
1.4 D
1.5 B
1.6 D

Question 2 – Waves

2.1.1 The number of complete waves passing a point ✓ in 1s ✓ (2)

2.1.2 $V = f\lambda$ ✓ $340 = \text{f} \times 8{,}5 \times 10^{-3}$ (4)

$\therefore f = \frac{340}{8{,}5\times10^{-3}\checkmark} = 40\ 000\checkmark\ \text{Hz}$

$= 4 \times 10^{4} Hz\checkmark$

2.1.3 Ultrasound. (1)

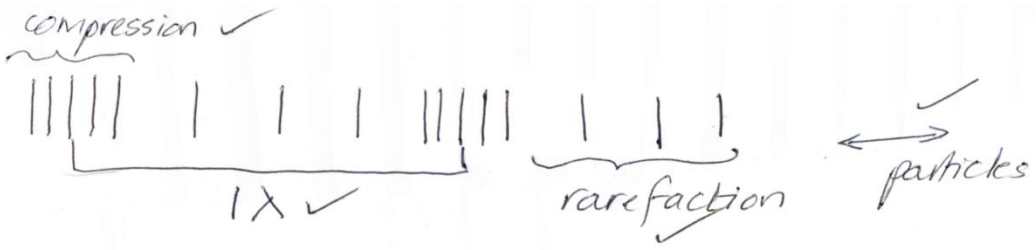
2.1.4 $\Delta x = v \times t\checkmark$ (3)

$= 340 \times \left(\frac{0{,}12}{2}\right)\checkmark$

$= 20{,}4m\ \checkmark$

2.1.5 Higher pitch ✓ (1)

2.1.6 (4)



2.2 Transverse waves / all travel at $3 \times 10^{8} m \cdot s^{-1}$ / can travel in a vacuum ✓. (1)

2.2.2 i) Infra-red (1)

ii) X-ray (1)

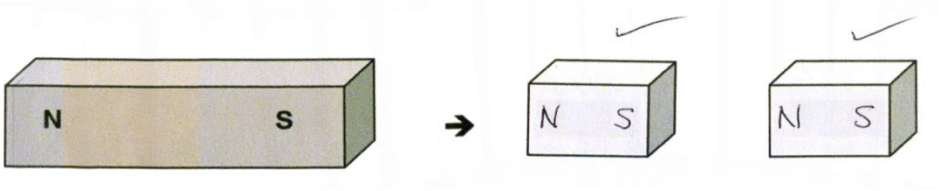
2.2.3 $E = hf = \frac{hc}{\lambda}$ ✓ $= \frac{6{,}6\times10^{-34}\times3\times10^{8}\checkmark}{700\times10^{-9}\checkmark(conv)}$ (4)

$$= 2{,}83 \times 10^{-19}\text{J} \checkmark$$ **[22]**

Question 3 Magnetism

3.1.1 A region in space where a magnet ✓ experiences a force of attraction or repulsion ✓. (2)

3.1.2 RIGHT to LEFT ✓ (1)

3.1.3 

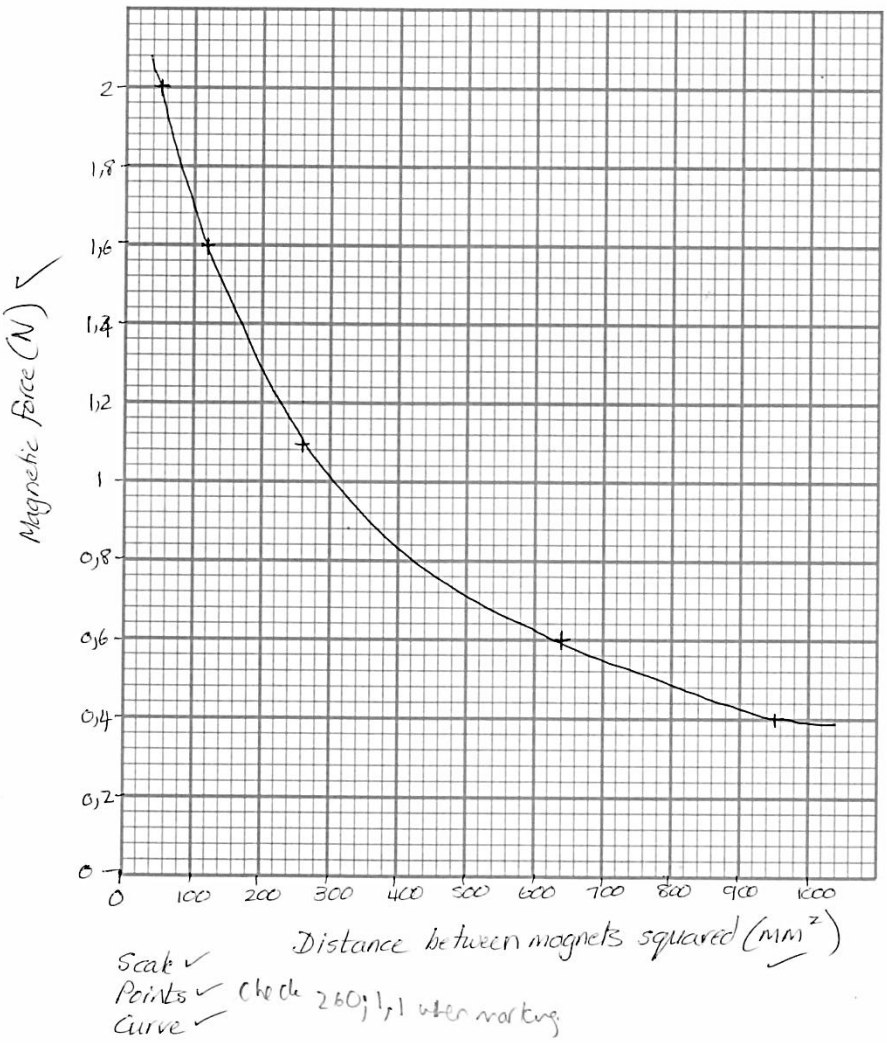
(2)

3.2.1 Magnetic force ✓ (1)

3.2.2 Same strength magnets ✓ / same magnetic orientation (N and S). (1)

3.2.3 (5)

3.2.4 Magnetic



force ✓ is inversely proportional ✓ to the distance between the magnets squared ✓. (3)

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Question 4 Electrostatics

4.1.1 Negative ✓ (1)

4.1.2 The child's hair lost electrons ✓ which the balloon gained ✓. (2)

4.1.3 $Q = nqe$ ✓ (4)

$= 300 \times 1{,}6 \times 10^{-19} C$ ✓

$\therefore Balloon\ has\ a\ charge\ of\ -4{,}8\ \times 10^{-17}C.$ ✓
$Hair\ has\ charge\ of\ +4{,}8 \times 10^{-17}\ C$ ✓

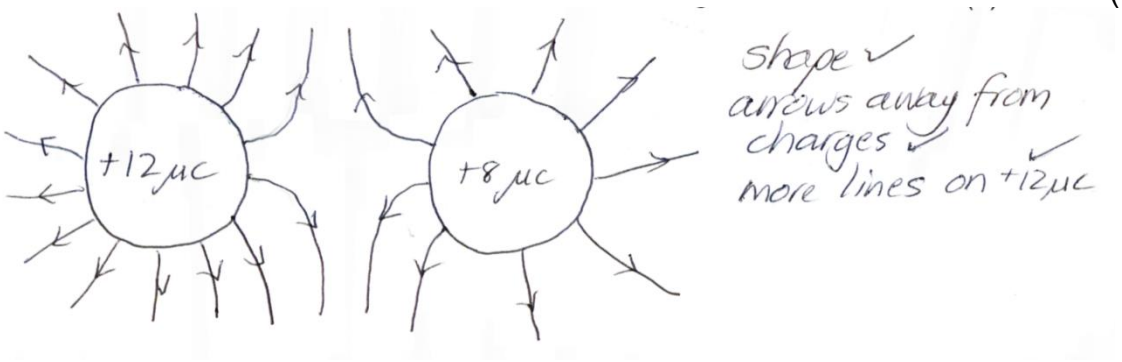
Mark $c.o.e$ according to 4.1.1

4.2 Quantisation of charge ✓ (1)

4.2.2 $Q = nqe \therefore n = \frac{Q}{qe} = \frac{-8\times10^{-9}C\ \checkmark(conv)}{-1{,}6\times10^{-19}C\ \checkmark}$ (3)

$= 5 \times 10^{10} e^{-}s$ ✓

4.3.1 (3)



4.3.2 Q to P ✓ (1)

4.3.3 Calculate the new charge on the spheres. (3)

$Q_{new} = \frac{Q_1 + Q_2}{2} = \frac{(+12) + (+8)}{2}$ ✓ ✓

$= 10\,\mu C$ ✓ / $10 \times 10^{-6}\,C$ / $1 \times 10^{-5}\,C$

4.3.4 How many electrons are transferred? (3)

$\frac{Q_{old} - Q_{new}}{-1,6 \times 10^{-19}}$

$= \frac{+8 \times 10^{-6} - 10 \times 10^{-6}}{-1,6 \times 10^{-19}}$ ✓ ✓

$= 1,25 \times 10^{13}$ e's ✓

$\Delta Q = 2 \times 10^{-6}$

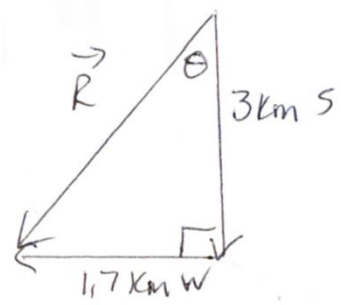
or $\frac{(12 - 10 \times 10^{-6})}{2}$

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Question 5 Kinematics

5.1.1 A change in position ✓ (2)

5.1.2 (4)



$\underset{R}{\rightarrow} = \sqrt{3^2 + 1,7^2}$ ✓

$= 3,45\ km$ ✓

$\tan\theta = \frac{1,7}{3}$

$\therefore \theta = 29,54°$ ✓

$\therefore a\ bearing\ of\ 180 + 29,54°\ (m)$

$= 209,54°$ ✓

(They may give angle relative to vertical or horizontal vector not on a bearing)

5.1.3 The length of path travelled ✓✓ (2)

5.1.4 $(3 \times 2) + (1{,}7 \times 2) = 9{,}4\ km$ ✓ $= 9400m$ ✓ (4)

$Av\ speed = \frac{9400}{(50 \times 60)}$ ✓

$= 3{,}13m \cdot s^{-1}$ ✓

5.2.1 $1s$ ✓ (1)

5.2.2 $Distance = Area\ under\ graph$ (3)

$= \frac{1}{2} b \times h$

$= \frac{1}{2} 3{,}5$ ✓ $\times 35$ ✓

$= 61{,}25m$ ✓

5.2.3 $Vf = V_i + a\Delta t$ ✓ (4)

✓$0 = 35 + a \cdot 3{,}5$

$= \frac{-35}{3{,}5}$

$= -10m.s^{-2}$

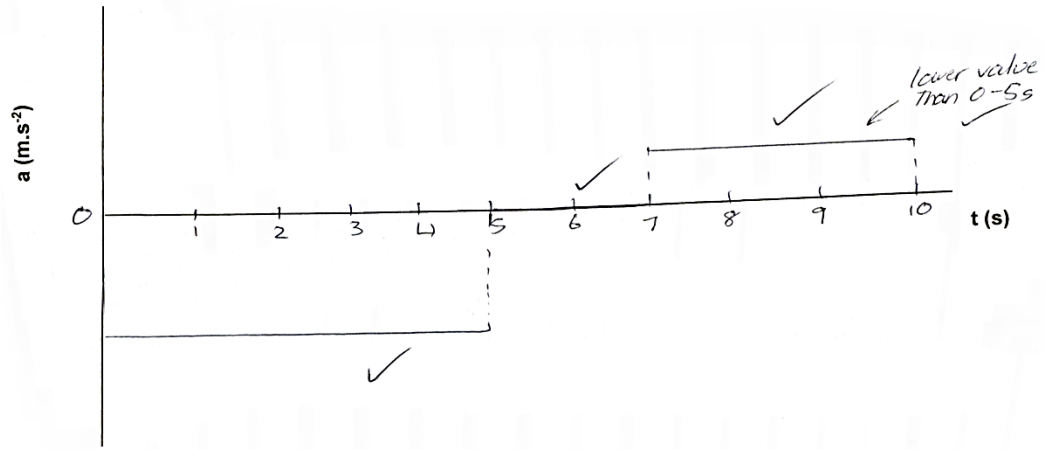
$ie\ 10m \cdot s^{-2}$ ✓$West$ ✓

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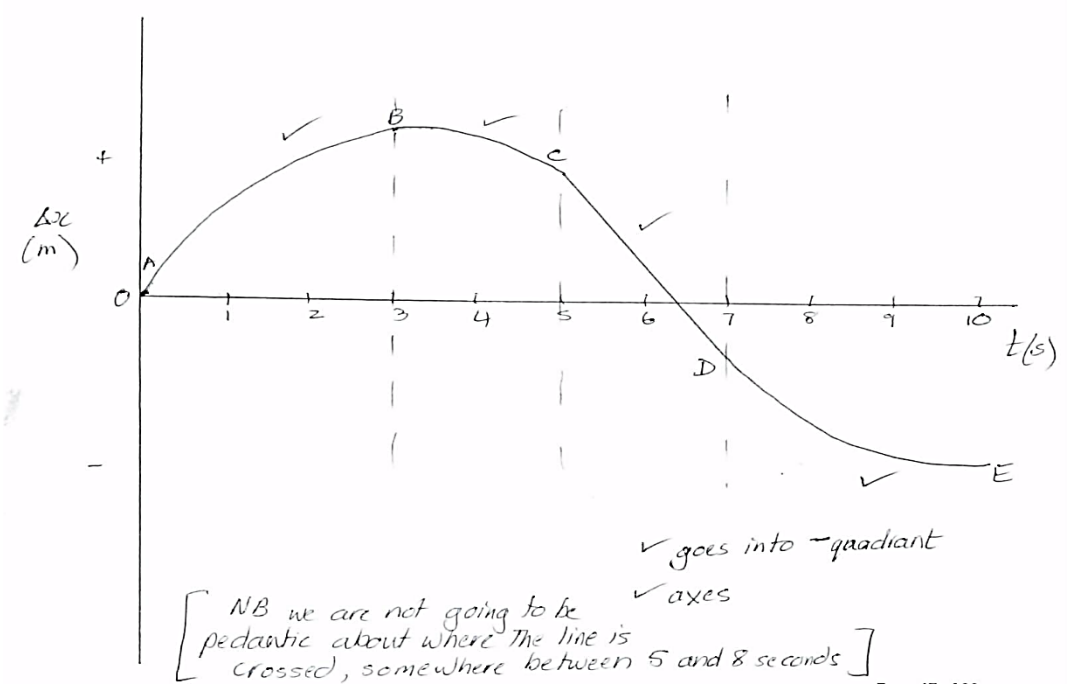
Question 6

6.1.1 Decreasing velocity ✓ south ✓ uniform / constant ✓ acceleration North ✓. (4)

6.1.2 (4)



6.1.3 (6)



6.2 *Distance travelled in* $8s$ $\therefore$ (4)

Truck

$\Delta x = v \times t$

$= 7{,}8 \times 8$

$= 62{,}4\ m$ ✓

Car

$\Delta x = v_i \Delta t + \frac{1}{2} a \Delta t^2$

$= 0 + \frac{1}{2} 4{,}2 \times 8^2$

$= 134{,}4\ m$ ✓

$134{,}4 - 62{,}4m\ (M)$ ✓

$= 72m$ ✓

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Question 7 Electric Circuits

7.1.1 $R_T = \frac{V_T \checkmark}{I} = \frac{3\times3 \checkmark}{3} = 3\ \Omega \checkmark$ (3)

7.1.2 $R_{//} = \frac{Product}{sum} = \frac{5\times2}{5+2} \checkmark$ (2)
$= 1{,}43\ \Omega$

7.1.3 $R_i = 3\Omega - 1{,}43\Omega (coe\ from\ 7.1.2) \checkmark$ (2)
$= 1{,}57\ \Omega \checkmark$

7.1.4 $V = IR$ OR $V_{RI} = 1{,}57 \times 3$ (2)
$= 3 \times 1{,}43 \checkmark (coe\ from\ 7.1.2)$ $= 4{,}71$
$= 4{,}29\ V \checkmark$ $v_{II} = (3 \times 3) - 4{,}71$
$\therefore can\ then\ use\ I = {}^{V}/_{R}$ $= 4{,}29\ V$

7.1.5 $R_{5\Omega}: R_{2\Omega}$ OR $I = \frac{V}{R} = \frac{4{,}29}{2} = 2{,}14\ A$ (3)
$5:2 \checkmark$
$\checkmark \therefore I_{5\Omega}: I_{2\Omega}$
$2:5\ (\therefore 7\ parts\ current)$
$\therefore \frac{5}{7} \times 3\ A = 2{,}14\ A \checkmark$

7.1.6 $W = VQ \checkmark$ (3)
$= 4{,}29V\ (coe\ from\ 7.1.4) \times 5 \checkmark$
$= 21{,}45\ J \checkmark$

7.2 E$= p \times T$ (4)
$= 3\ kW \checkmark \times 0{,}75\ hours \checkmark$
$= 2{,}25\ kWh \checkmark$
$2{,}25 \times R3{,}20 = R7{,}20 \checkmark$

7.3.1 Same length ✓
Same diameter / thickness ✓ (2)

7.3.2 **Wire A** **Wire B** (5)

$Gradient = \frac{5{,}6-3{,}8 \checkmark}{0{,}5-0{,}34 \checkmark}$ $\frac{4-2}{0{,}7-0{,}36 \checkmark}$

$= 11{,}25$ ✓Ω $= 5{,}88$ ✓Ω

NB cannot use points given. Choose own points on line and indicate on graph.
MUST have a CHANGE in y and a CHANGE in x.

7.4 Wire A ✓ (1)

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